

Homework 12 - More filters & Fourier- & Wavelet analysis

Deadline: 4th test date (4th of June)

The mandatory questions account for 100 points, while the optional questions together contribute up to an additional 20 points.

1. Implement a Python script to filter white noise with a bandpass FIR filter and compare spectra.

2. Load the data from the `eeg.bin` file, which contains a continuous sequence of 32-bit floating-point EEG samples.

Determine the sampling frequency of the recording.

Hint: EEG recordings often contain a narrow spectral peak caused by power-line interference.

3.  Load the EEG signal from the `eeg.bin` file and apply both a low-pass FIR filter and a low-pass IIR filter with a cutoff frequency of 2 Hz.

Demonstrate the effect of phase shift in the outputs of the two filters and compare their phase characteristics.

Then apply forward-backward filtering and show how it mitigates phase distortion in both cases.

4. Describe the purpose of the Short Time Fourier Transform (STFT).
5. What is the Heisenberg Uncertainty Principle in the context of signal processing?
6. Write up the continuous wavelet transform (CWT) formula and explain its components.
7. Explain the relationship between the Continuous Wavelet Transform (CWT) and bandpass filtering.

8. Explain the concept of vanishing moments in wavelet theory.

9.  Show that the admissibility criterion implies the zero-mean condition.

10. What is the main purpose of wavelet transform in signal processing?

11. Define the term "scaling function" in the context of wavelet transforms.

12. What is the difference between the Continuous Wavelet Transform (CWT) and the Discrete Wavelet Transform (DWT)?

13. For the Haar scaling function

$$\phi(t) = \begin{cases} 1, & 0 \leq t < 1, \\ 0, & \text{otherwise,} \end{cases}$$

show that the two-scale relation

$$\phi(t) = \phi(2t) + \phi(2t - 1)$$

is satisfied. Then, using

$$\psi(t) = \phi(2t) - \phi(2t - 1),$$

derive the Haar wavelet function explicitly.